Local Mean-Force and Harmonic/Manifold Refinement Specification

Stage 11E3

mdstats

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Contents

1 Scope **1**

2 Borrowed methods and package-specific constructions **2**

3 Coordinate and force contract **2**

4 Matched-kernel conditional mean force **2**

5 Density-score comparison **3**

6 Lifted residence chart **3**

7 Symmetric local force fit **3**

8 Curvature classes **3**

9 Residence covariance diagnostic **4**

10 Evidence status **4**

11 Resource and serialization contract **4**

12 Non-goals **4**

13 Acceptance tests **5**

14 References **5**

1 Scope

Stage 11E3 refines the position-derived statistical candidates from Stage 11E2 with force evidence from the Stage 11E0b sample catalog. It does not create, delete, merge, or label spatial sites. A candidate remains present when force data are unavailable, inadmissible for PMF use, locally under-sampled, singular, or inconsistent with a stable harmonic model.

The implementation owner is:

```
mdstats.analysis.density.force_refinement
```

The stage consumes one source-compatible triple:

```
FrameworkAlignedIonSampleCatalog
PeriodicSpeciesDensityEstimate
DensityAttractorCatalog
```

and returns one immutable `ForceRefinementCatalog`.

2 Borrowed methods and package-specific constructions

The conditional-average-force interpretation follows Darve and Pohorille (2001), DOI 10.1063/1.1410978. The local linear force-matching interpretation follows Noid et al. (2008), DOI 10.1063/1.2938860. Complete-system blocking for correlated uncertainty follows Flyvbjerg and Petersen (1989), DOI 10.1063/1.457480.

The following are mdstats-specific constructions:

- exact source binding across E0b, E1, and E2 signatures;
- retaining every E2 attractor regardless of force-evidence status;
- matched use of the E1 periodized Gaussian and image-truncation certificate;
- explicit Cartesian-to-fractional covector conversion before score comparison;
- support intersection with the E1 support mask;
- lifted periodic residence charts bound to one E2 attractor;
- separate density anchor, force center, curvature class, and chart-containment status;
- separate ordinary fit, covariance diagnostic, and density-force residual; and
- fail-closed serialization and resource preflight.

3 Coordinate and force contract

Let the registered periodic domain use row-vector fractional coordinates $\mathbf{q}$ and cell matrix H:

$$\mathbf{x} = \mathbf{q}H.$$

A registered Cartesian force covector $\mathbf{F}_x$ is transformed to the fractional coordinate measure by work invariance:

$$\mathbf{F}_q = \mathbf{F}_x H^\top, \qquad \mathbf{F}_x\, d\mathbf{x}^\top = \mathbf{F}_q\, d\mathbf{q}^\top.$$

The E1 density score $d_q \log p$ and $\mathbf{F}_q$ are therefore compared in the same covector measure. The analysis metric is not applied until an orthonormal local chart is required.

If $G = LL^\top$ is the E1 analysis metric and $\delta_y = \delta_q L$, then local force covectors transform as

$$\mathbf{F}_y = \mathbf{F}_q L^{-\top}.$$

4 Matched-kernel conditional mean force

Only the E0b `pmf_force` evidence subset is admissible. For query node $\mathbf{q}_g$, represented-time weights w_n, forces $\mathbf{F}_{q,n}$, and the same periodized Gaussian $K_h^{\mathbb{T}^3}$ used by E1,

$$\overline{\mathbf{F}}_q(\mathbf{q}_g) = \frac{\sum_n w_n K_h^{\mathbb{T}^3}(\mathbf{q}_g - \mathbf{q}_n)\mathbf{F}_{q,n}}{\sum_n w_n K_h^{\mathbb{T}^3}(\mathbf{q}_g - \mathbf{q}_n)}.$$

The field stores:

- conditional force covector;
- conditional force covariance;
- local effective sample size;
- support mask;
- optional complete-system block standard error;
- represented ion time and force-sample count; and
- E0b/E1 source signatures.

A node is force-supported only when it also belongs to the E1 support mask and passes the configured local effective-sample threshold. Unsupported field values are finite zeros; the support mask is authoritative.

5 Density-score comparison

At declared constant temperature T and for PMF-admissible equilibrium sampling,

$$\overline{\mathbf{F}}_q(\mathbf{q}) \approx k_\mathrm{B}T\, d_q \log p(\mathbf{q}).$$

For each E2 attractor, the result records the norm of the residual at the representative logical node when both channels are supported. This residual is a diagnostic. It does not overwrite either field or project the force field onto a conservative subspace.

6 Lifted residence chart

Samples assigned to attractor i use the E2 basin owner at the nearest periodic logical node. Relative positions are lifted by

$$\delta_q = (\mathbf{q} - \mathbf{q}_i) - \mathrm{rint}(\mathbf{q} - \mathbf{q}_i).$$

A lift is accepted only when every component lies strictly inside the periodic half-cell cut. The E3 fit chart is the union of the E2 local chart and accepted residence lifts. This extension is recorded by the fit sample indices; it does not alter the E2 topology.

7 Symmetric local force fit

For point candidates and resolved extended candidates, fit

$$\mathbf{f}_n = \mathbf{b}_i - \delta_{y,n} K_i + \epsilon_n, \qquad K_i = K_i^\mathsf{T},$$

with represented-time weighted least squares. The nine fitted parameters are three intercept components and six independent stiffness components. The result retains:

- design rank and condition number;
- intercept and symmetric stiffness;
- stiffness eigenvalues and eigenvectors;
- residual covariance;
- parameter standard errors;
- force-defined point center where identifiable; and
- center-within-chart status.

For nonsingular point fits,

$$\delta_y^{(F)} = \mathbf{b}_i K_i^{-1}.$$

The center is not imposed for an extended attractor.

8 Curvature classes

The force fit reports one of:

```
stable_point
saddle_or_unstable
soft_manifold
flat_or_unresolved
not_evaluated
```

A stable point requires all stiffness eigenvalues above the declared minimum. A saddle or unstable fit has at least one eigenvalue below the negative threshold. A resolved one-dimensional E2 manifold is `soft_manifold` when two normal directions are restoring and the soft eigenvalue is small relative to the largest stiffness. Otherwise the result remains unresolved.

9 Residence covariance diagnostic

For all position-admissible samples assigned to the provisional residence basin, compute the represented-time weighted covariance in the same orthonormal chart:

$$\Sigma_i^{(\mathrm{res})} = \mathrm{Cov}(\delta_y \mid B_i).$$

When K_i is positive definite and a constant temperature is declared, also report

$$\Sigma_i^{(\mathrm{harm})} = k_{\mathrm{B}} T K_i^{-1}$$

and the relative Frobenius discrepancy. This is diagnostic only. Basin truncation, anharmonicity, many-ion conditioning, and finite sampling may make the two covariances disagree.

10 Evidence status

Every E2 attractor receives exactly one `LocalForceRefinement`. Its independent force-evidence status is one of:

```
resolved
force_unavailable
pmf_provenance_rejected
insufficient_local_support
chart_unresolved
rank_deficient
ill_conditioned
center_outside_chart
```

No non-resolved status deletes the position-derived candidate.

11 Resource and serialization contract

`LocalMeanForceResourcePolicy` preflights grid nodes, force samples, Gaussian image terms, workspace, output bytes, and attractor count before allocation. All options, fields, refinements, and catalogs have deterministic SHA-256 signatures and strict replay constructors. Field replay requires numerical values; summary-only dictionaries are not replayable scientific objects.

12 Non-goals

Stage 11E3 does not:

- reconstruct a global PMF;
- project a raw force field to its conservative component;
- perform Helmholtz-Hodge decomposition;
- infer transition events or residences;

- assign structural or crystallographic site labels;
- estimate barriers or rates; or
- replace the E2 nonparametric basin catalog with harmonic ellipsoids.

13 Acceptance tests

The focused gate must verify:

1. a synthetic harmonic well recovers center and stiffness;
2. unstable curvature is not labeled a stable point;
3. an extended attractor preserves a soft direction without a point center;
4. missing or PMF-inadmissible force data preserve all spatial attractors;
5. insufficient samples and singular designs lower evidence status;
6. matched fields retain E1 support and block uncertainty;
7. source mismatch and resource excess fail closed;
8. serialization replay is exact and tamper-evident; and
9. public exports and stage metadata are stable.

14 References

Darve, E., and Pohorille, A. (2001). *Calculating Free Energies Using Average Force.* Journal of Chemical Physics 115, 9169-9183. DOI: 10.1063/1.1410978.

Noid, W. G., et al. (2008). *The Multiscale Coarse-Graining Method. I. A Rigorous Bridge between Atomistic and Coarse-Grained Models.* Journal of Chemical Physics 128, 244114. DOI: 10.1063/1.2938860.

Flyvbjerg, H., and Petersen, H. G. (1989). *Error Estimates on Averages of Correlated Data.* Journal of Chemical Physics 91, 461-466. DOI: 10.1063/1.457480.